Wednesday, 1/21

→ is this $O(h^3)$ local for explicit

→ to check compute

$$|y_{i+1} - w_{i-1}| \overset{?}{=} O(h^?)$$

Expand: $y_{i+1} = y_i + h\, \underset{f(t_i, y_i)}{\underset{\shortparallel}{y_i'}} + \frac{h^2}{2} \underset{\frac{d}{dt}[f(t_i, y_i)]}{\underset{\shortparallel}{y_i''}} + O(h^3)$

Note:

idea: taylor expand $f(t_i + h,\ w_i + hf(t_i, y_i))$

Recall - Taylor Expansion - single var

$$f(x) = f(x_0) + (x - x_0) f'(x_0) + \frac{(x-x_0)^2}{2!} f''(x_0) + \dots$$

2-variable Taylor Exp.

$$f(x,y) = f(x_0, y_0) + (x - x_0) f_x(x_0, y_0) + (y - y_0) f_y(x_0, y_0) + \Big\} \text{ first order}$$

coeff from pascals △ $\quad \frac{(x-x_0)^2}{2!} f_{xx}(x_0, y_0) + \cancel{2}\frac{(x-x_0)(y-y_0)}{\cancel{2!}} f_{xy}(x_0, y_0) + \frac{(y-y_0)^2}{2!} f_{yy}(x_0, y_0) \dots \Big\}$ second order

Now, let's apply to explicit trapezoid error..

$$f(\overbrace{t_i + h}^{t},\ \overbrace{w_i + hf(t_i, w_i)}^{w}) + O(h^2)$$

$$= f(t_i, w_i) + h f_t(t_i, w_i) + h f(t_i, w_i) f_w(t_i, w_i) + O(h^2)$$

↕ Compare w_{i+1} & y_{i+1}

$$y_{i+1} = y_i + hf(t_i, w_i) + \frac{h^2}{2}\Big[f_t(t, y) + f_y(t, y) f(t, y)\Big]_{t = t_i}$$

... on your own, you can cancel everything out to $O(h^3)$ for the local error

Global Error

- Euler: $O(h)$ — order 1
- Im. Trap.: $O(h^2)$ — order 2
- Ex. Trap.: $O(h^2)$ — order 2

Existence & Uniqueness of Solutions

→ Before approximating a soln. to $y' = f$. $y(t_0) = y_0$, we should ask if a soln. exists

→ this existence depends on f

Thm If f satisfies the lipschitz condition

$$|f(t, y_2) - f(t, y_1)| < L|y_2 - y_2|$$

* note: $t \in$ domain of y

where $L > 0$ and finite, then $\exists!$ soln to IVP

→ looks like a quotient; weak req. for differentiabilty

→ this req. is weaker than differentiability

→ for us, we assume f is differentiable, moreover it has a Taylor Exp. for some non-zero radius of conv.

Friday Jan 23

def a numerical method for solving / approx solns to ODEs is called order k if it's global error is $O(h^k)$

→ How do we generate more accurate methods?

one idea: Start where Euler stopped – Taylor Method

$$y_{i+1} = y_i + hy_i' + \frac{h^2}{2}y_i'' + O(h^3) \rightarrow \text{in terms of } f(t,y)$$

$$y_{i+1} = y_i + h\underbrace{f_i}_{f(t_i,w_i)} + \frac{h^2}{2}\frac{d}{dt}[f(t,y)]\Big|_{t=t_i} + \frac{h^3}{3!}\frac{d^2}{dt}[f(t,y)]\Big|_{t=t_i} + O(h^4)$$

$\frac{dy}{dt} = f(t,y)$

$$\rightarrow \frac{d}{dt}[f(t,y(t))] = f_t(t,y) + f_y(t,y)f(t,y)$$

$$\Rightarrow \frac{d^2}{dt^3}\ (f_{tt}(t,y) + f_{ty}(t,y)f(t,y)) + f(t,y)[f_{ty}(t,y) + f_{yy}(t,y)f(t,y)] + f_y(t,y)[f_t(t,y) + f_y(t,y)f(t,y)]$$

Ex Determine the 2nd-order Taylor Method for $y' = ty + t^3 = f(t,y)$

$$\frac{d}{dt}(ty+t^3) = y + 3t^2 + t(ty+t^3) =$$

$$\frac{d^2}{dt^2} = (6t + (1)(ty+t^3)) + (ty+t^3) + (t)[y + 3t^2 + t(ty+t^3)]$$

$$w_{i+1} = w_i + h(t_iw_i + t_i^3) + \frac{h^2}{2}(w_i + 3t_i^2 + t_i(t_iw_i + t_i^3))$$

the taylor method of order k can be expressed as:

$$w_{i+1} = w_i + \underbrace{h\left[f_i + \frac{h}{2}\frac{d}{dt}[f] + \frac{h^2}{3!}\frac{d^2}{dt^2}[f] + \dots + \frac{h^{n-1}}{n!}\frac{d^{n-1}}{dt^{(n-1)}}[f]\right]}_{f(t_i,y_i) \text{ or } T^n}$$

How good is a numerical method?

1) Consistency
2) Convergence
3) Stability wrt Round off

1) Consistency:

note: our methods are "difference eq" ie $y_{i+1} = y_i + h f_i$ & mimicking the differential eq

→ the difference eq. generates solutions that should approx to the diff. equation

do the differential & difference eq "line up"

→ Taylor: $y_{i+1} = y_i + h\varphi(t_i, y_i)$

$$\rightarrow \quad \frac{y_{i+1} - y_i}{h} - \varphi(t_i, y_i) = 0$$

a method is consistent if

$$\lim_{h \to 0} \tau_{i+1}(h)$$

$$\lim_{h \to 0} \frac{y_{i+1} - y_i}{h} - f(t_i, y_i) = 0$$